Avishkar Pawar

Academic CV

GitHub | LinkedIn | Portfolio | avishkarpawar004@gmail.com | +91-8623005430 | Pune, India

Summary

Final-year B.Tech. Computer Engineering student at MIT Academy of Engineering, Pune with 9.24 CGPA. GATE 2026 qualified with AIR 3124 and GATE Score 601. Academic and project work focuses on distributed systems, database internals, backend engineering, and AI/ML infrastructure. Completed Software Developer internships at OSMOS and Core-Decimal Solutions.

Work Experience

Software Developer Intern, OSMOS Jul 2025 - May 2026

- Developed backend services using Node.js and Python for application and product development.
- Contributed to product development from scratch by designing, implementing, testing, and improving core features.

Software Developer Intern, Core-Decimal Solutions Feb 2025 - Aug 2025

- Built full-stack features using Vue.js, Express.js, Sequelize, and MySQL in an MVC architecture.
- Developed reusable Vue.js components, integrated RESTful APIs with Axios, and designed Sequelize data mappings.

Software Development Intern, Campus Credential 2024

- Worked on web application features, database-backed workflows, and production-oriented bug fixes.

Projects

MiniDB - Distributed Key-Value Database | *C++17, Boost.Asio, Protocol Buffers, CMake* GitHub

- Built B+ Tree indexing, MVCC transactions, Raft consensus, SQL-like query processing, sharding, TCP networking, and HTTP monitoring for 10,000+ operations/sec with sub-millisecond local latency.

StreamFlow - Kafka-like Event Streaming Platform | *Java 17, Spring Boot, Netty, Docker* GitHub

- Implemented partitions, leader-follower replication, ISR tracking, consumer groups, log-based storage, zero-copy transfer, admin APIs, and monitoring hooks for high-throughput messaging.

VectorFlow - AI Vector Database | *Python, FastAPI, NumPy, Numba, LangChain* GitHub

- Built HNSW nearest-neighbor search, WAL persistence, distributed sharding, REST APIs, optimized distance computation, and RAG pipeline integration for semantic search.

Player Re-Identification in Sports Footage | *Python, computer vision, deep learning* GitHub

- Built a system for re-identifying players across camera angles in sports footage using visual features and tracking-oriented matching.

Education

B.Tech. in Computer Engineering, MIT Academy of Engineering, Pune 2022 - 2026

CGPA: 9.24/10

Achievements

- GATE 2026 Qualified: AIR 3124, GATE Score 601, 51.45/100 marks in Computer Science.
- GATE 2025 Qualified: AIR 9860, GATE Score 453, 39.32/100 marks.
- First Place, Technodium 25 technical competition.
- Certifications: AWS Cloud Practitioner, CCNA, Python Essentials, PCAP, NPTEL Modern C++, NPTEL Machine Learning, NPTEL DSA, NPTEL Cloud Computing, NPTEL Cryptography, Saylor Cryptography, Red Hat Academy.

Skills

Focus Areas	Computer Engineering, GATE CS, Distributed Systems, Database Systems, AI/ML
Languages	C++, Java, Python, JavaScript, TypeScript, SQL, PHP
Frameworks	Spring Boot, FastAPI, React, Node.js, Express.js, Vue.js, Flask
Databases	PostgreSQL, MySQL, MongoDB, Redis, custom database implementations
AI/ML	TensorFlow, PyTorch, OpenAI API, LangChain, NumPy, Numba, computer vision
Tools	Git, Docker, Linux, AWS, CMake, Maven, Prometheus, REST APIs
Core CS	Data Structures and Algorithms, Operating Systems, Database Systems, Computer Networks, Distributed Systems, System Design